Python Engineering - Class 06 Reference Guide

Dictionaries and Sets

Reference guide to the syntax and tools used in this class. It explains what each element does, how it is written and when it is useful.

Quick map

Concept	Syntax	Idea
dict	config={'g':9.81}	Mapping from unique keys to values.
key lookup	config['g']	Retrieve a value by key.
get()	config.get('g')	Retrieve with optional default rather than direct KeyError.
keys()	config.keys()	View dictionary keys.
values()	config.values()	View stored values.
items()	config.items()	Iterate over key-value pairs.
membership	'g' in config	Check whether a key exists.
set	required={'g','dt'}	Unordered collection of unique hashable values.
set difference	required - provided	Find required items that are missing.
set intersection	A & B	Find elements present in both sets.

1. dict

Syntax: `config={'g':9.81}`

Mapping from unique keys to values.

```
config = {'g': 9.81, 'dt': 0.1}
```

Control question: before running the example, predict the result type and shape.

2. key lookup

Syntax: `config['g']`

Retrieve a value by key.

```
g = config['g']
```

Control question: before running the example, predict the result type and shape.

3. get()

Syntax: `config.get('g')`

Retrieve with optional default rather than direct KeyError.

```
g = config.get('g', 9.81)
```

Control question: before running the example, predict the result type and shape.

4. keys()

Syntax: `config.keys()`

View dictionary keys.

```
provided = set(config.keys())
```

Control question: before running the example, predict the result type and shape.

5. values()

Syntax: `config.values()`

View stored values.

```
for value in config.values():
    ...
```

Control question: before running the example, predict the result type and shape.

6. items()

Syntax: `config.items()`

Iterate over key-value pairs.

```
for key, value in config.items():
    ...
```

Control question: before running the example, predict the result type and shape.

7. membership

Syntax: `'g' in config`

Check whether a key exists.

```
if 'g' in config:
    ...
```

Control question: before running the example, predict the result type and shape.

8. set

Syntax: `required={'g','dt'}`

Unordered collection of unique hashable values.

```
required = {'h0', 'v0', 'g', 'dt'}
```

Control question: before running the example, predict the result type and shape.

9. set difference

Syntax: `required - provided`

Find required items that are missing.

```
missing = required - provided
```

Control question: before running the example, predict the result type and shape.

10. set intersection

Syntax: `A & B`

Find elements present in both sets.

```
common = required & provided
```

Control question: before running the example, predict the result type and shape.

Common mistakes

- Confusing a function name with a function call using parentheses.
- Not checking types, indices or shapes before an operation.
- Using a tool because it works without being able to explain what it receives and returns.

Mental cheat sheet

```
dict: config={'g':9.81}
key lookup: config['g']
get(): config.get('g')
keys(): config.keys()
values(): config.values()
items(): config.items()
membership: 'g' in config
set: required={'g','dt'}
set difference: required - provided
set intersection: A & B
```